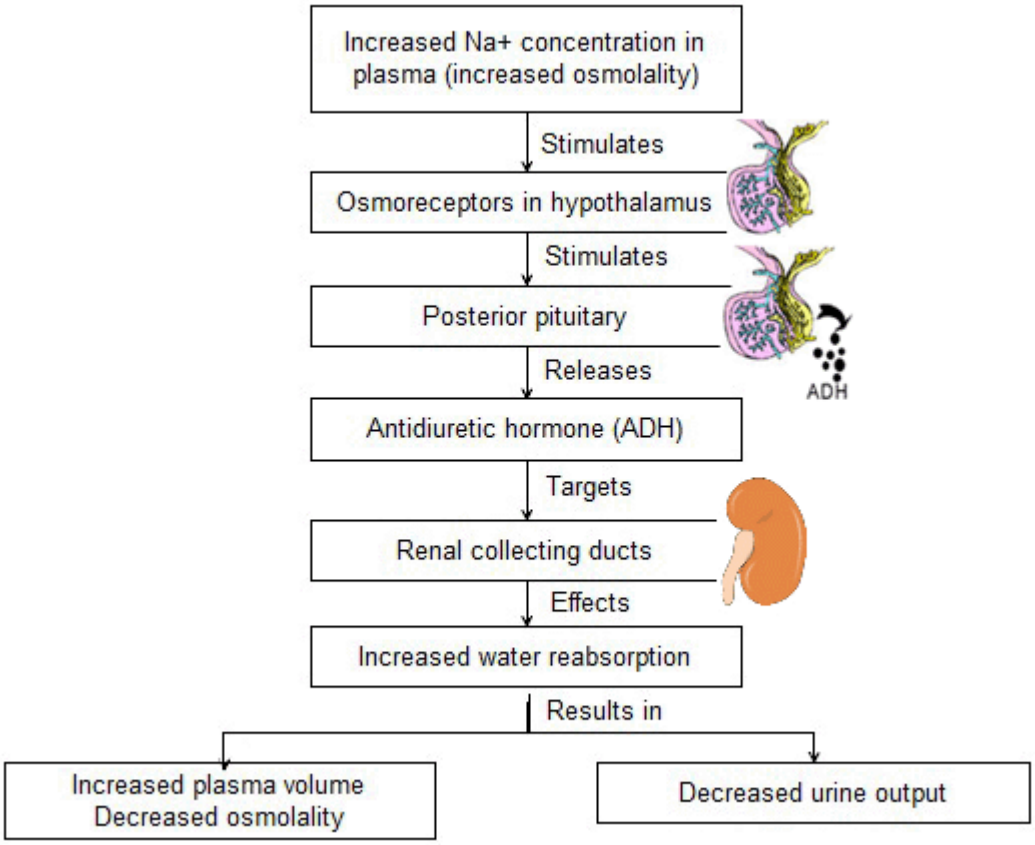
Antidiuretic hormone (vasopressin) promotes the reabsorption of free water. It is released from the posterior pituitary gland in response to increased serum osmolality and targets the collecting ducts of the kidneys.

The osmoreceptors in the posterior pituitary are sensitive to as little a change in the serum osmolality as 1%. Stretch receptors in the left atrium are sensitive to changes in blood volume and are also capable of releasing ADH.

Conditions associated with stimulation of the release of vasopressin include: contracted extracellular volume, hypotension, hypernatremia, nausea and vomiting, congestive heart failure, cirrhosis, hypothyroidism, and the increased presence of angiotensin II, histamine, bradykinin, and catecholamines.

Other actions of ADH include promotion of hemostasis by increasing the circulating levels of von Willebrand factor and factor VIII and increasing blood pressure by constricting smoother muscle in the vascular system, primarily the splanchnic, renal, and coronary systems. Because of the effect on coronary vasculature, it can produce myocardial ischemia in susceptible patients and must be administered with caution.



Nagelhout JJ, Plaus K. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 845-847.

Copyright 2015 Prodigy Anesthesia, Inc. It is unlawful to copy, distribute, and/or sell this material.